

Actual Causes of Death in the United States, 2000

Ali H. Mokdad, PhD

James S. Marks, MD, MPH

Donna F. Stroup, PhD, MSc

Julie L. Gerberding, MD, MPH

IN A SEMINAL 1993 ARTICLE, McGinnis and Foege¹ described the major external (nongenetic) modifiable factors that contributed to death in the United States and labeled them the “actual causes of death.” During the 1990s, substantial lifestyle pattern changes may have led to variations in actual causes of death. Mortality rates from heart disease, stroke, and cancer have declined.² At the same time, behavioral changes have led to an increased prevalence of obesity and diabetes.³

Most diseases and injuries have multiple potential causes and several factors and conditions may contribute to a single death. Therefore, it is a challenge to estimate the contribution of each factor to mortality. In this article, we used published causes of death reported to the Centers for Disease Control and Prevention (CDC) for 2000, relative risks (RRs), and prevalence estimates from published literature and governmental reports to update actual causes of death in the United States—a method similar to that used by McGinnis and Foege.

METHODS

Our literature review used a MEDLINE database search of English-language articles that identified epidemiological, clinical, and laboratory studies linking risk behaviors and mortality. Our search criteria were to include all ar-

Context Modifiable behavioral risk factors are leading causes of mortality in the United States. Quantifying these will provide insight into the effects of recent trends and the implications of missed prevention opportunities.

Objectives To identify and quantify the leading causes of mortality in the United States.

Design Comprehensive MEDLINE search of English-language articles that identified epidemiological, clinical, and laboratory studies linking risk behaviors and mortality. The search was initially restricted to articles published during or after 1990, but we later included relevant articles published in 1980 to December 31, 2002. Prevalence and relative risk were identified during the literature search. We used 2000 mortality data reported to the Centers for Disease Control and Prevention to identify the causes and number of deaths. The estimates of cause of death were computed by multiplying estimates of the cause-attributable fraction of preventable deaths with the total mortality data.

Main Outcome Measures Actual causes of death.

Results The leading causes of death in 2000 were tobacco (435 000 deaths; 18.1% of total US deaths), poor diet and physical inactivity (400 000 deaths; 16.6%), and alcohol consumption (85 000 deaths; 3.5%). Other actual causes of death were microbial agents (75 000), toxic agents (55 000), motor vehicle crashes (43 000), incidents involving firearms (29 000), sexual behaviors (20 000), and illicit use of drugs (17 000).

Conclusions These analyses show that smoking remains the leading cause of mortality. However, poor diet and physical inactivity may soon overtake tobacco as the leading cause of death. These findings, along with escalating health care costs and aging population, argue persuasively that the need to establish a more preventive orientation in the US health care and public health systems has become more urgent.

JAMA. 2004;291:1238-1245

www.jama.com

ticles including the following key words: *mortality, smoking, physical activity, diet, obesity, alcohol, microbial agents, toxic agents, motor vehicle, firearms, sexual behavior, illicit drug use*. Our search allowed for words with similar meaning to be included (ie, exercise as well as physical activity). The search was initially restricted to articles published during or after 1990, but we later included relevant articles published in 1980 to December 31, 2002 (search strategies are available from the authors on request). For each risk factor, we used the prevalence and RR identified by the literature search. To identify the causes and number of

deaths, we used mortality data reported in 2000 to the CDC.⁴ We used no unpublished information or data.

We used the following formula to calculate attributable fractions for each disease: $[(P_0 + \sum P_i (RR_i)) - 1] / [P_0 + \sum P_i (RR_i)]$, in which P_0 is the percentage of individuals in the United States not engaging in the risk behavior, P_i is the per-

Author Affiliations: Division of Adult and Community Health (Dr Mokdad), Office of the Director (Drs Marks and Stroup), National Center for Chronic Disease Prevention and Health Promotion and Office of the Director (Dr Gerberding), Centers for Disease Control and Prevention, Atlanta, Ga.

Corresponding Author: Ali H. Mokdad, PhD, Division of Adult and Community Health, 4770 Buford Hwy, NE, Mailstop K66, Atlanta, GA 30341 (amokdad@cdc.gov).

For editorial comment see p 1263.

centage engaging in separate categories of the risk behavior, and RR_i is the RR of death for each separate category relative to none. For instance, in the case of smoking, P_0 is the percentage of persons who never smoked, P_1 is the percentage of former smokers, P_2 is the percentage of current smokers, RR_1 is the RR of a certain type of death for former smokers compared with those who never smoked, and RR_2 is the RR of death for current smokers compared with those who never smoked. We then multiplied estimates of the cause-attributable fraction of preventable deaths by total mortality data. Whenever possible, we used RRs of death and mortality data by other variables such as age, sex, and race.

We estimated ranges for our estimated number of deaths by using the smallest and highest RRs and their boundaries when available. When data were available, we used specific underlying causes of death in deriving some of our estimates (ie, firearms, motor vehicles, and illicit drug use). Further details of these methods may vary due to availability of data and are presented in each section below. We used SAS (version 8.2, SAS Institute Inc, Cary, NC) and SUDAAN (version 8.0, Research Triangle Institute, Research Triangle Park, NC) statistical software.

RESULTS

The number of deaths in the United States in 2000 was 2.4 million, which is an increase of more than 250 000 deaths in comparison with the 1990 total, due largely to population growth and increasing age.^{2,4} Leading causes of death were diseases of the heart (710 760), malignant neoplasms (553 091), and cerebrovascular diseases (167 661) (TABLE 1).

Tobacco

We used methods and software used in previous CDC reports to compute the annual smoking-attributable mortality for 2000.^{5,6} As in previous reports, we used RRs for each cause of death from the American Cancer Society's Cancer

Table 1. Leading Causes of Death in the United States in 2000*

Cause of Death	No. of Deaths	Death Rate per 100 000 Population
Heart disease	710 760	258.2
Malignant neoplasm	553 091	200.9
Cerebrovascular disease	167 661	60.9
Chronic lower respiratory tract disease	122 009	44.3
Unintentional injuries	97 900	35.6
Diabetes mellitus	69 301	25.2
Influenza and pneumonia	65 313	23.7
Alzheimer disease	49 558	18
Nephritis, nephrotic syndrome, and nephrosis	37 251	13.5
Septicemia	31 224	11.3
Other	499 283	181.4
Total	2 403 351	873.1

*Data are from Minino et al.⁴

Prevention Study II⁷ and included deaths due to secondhand smoking.

We used data from the Behavioral Risk Factor Surveillance System (BRFSS), a cross-sectional telephone survey conducted by state health departments with the CDC's assistance, to determine changes in US smoking prevalence from 1995-1999 to 2000. A detailed description of survey methods is available elsewhere.⁸ A slight decline in smoking was observed from 1995-1999 to 2000. The prevalence of smoking in 1995-1999 was 22.8% for current smokers (males: 25.1%; females: 20.6%), 24.1% for former smokers (males: 28.3%; females: 20.3%), and 53.1% for never-smokers (males: 46.5%; females: 59.2%). In 2000, these estimates were 22.2% for current smokers (males: 24.1%; females: 20.5%), 24.4% for former smokers (males: 28.3%; females: 20.7%), and 53.4% for never-smokers (males: 47.6%; females: 58.8%).

We estimate that approximately 435 000 deaths were attributable to smoking in 2000, which is an increase of 35 000 deaths from 1990 (TABLE 2). This increase is due to the inclusion of 35 000 deaths due to secondhand smoking and 1000 infant deaths due to maternal smoking, which were not included in the article by McGinnis and Foege.¹

Poor Diet and Physical Inactivity

To assess the impact of poor diet and physical inactivity on mortality, we computed annual deaths due to over-

weight.⁹ Recent articles have reported that overweight increased in all segments of the US population.^{10,11} To derive the attributable number of deaths due to overweight, we used estimates from the CDC's 1999 and 2000 National Health and Nutrition Examination Surveys.¹² We used the same procedure reported by Allison et al¹³ to estimate annual overweight-attributable deaths. We used the body mass index (BMI) range of 23 to 25 as our reference category to match the method used by Allison et al. Body mass index is calculated as weight in kilograms divided by the square of the height in meters. Using data from the 1999 and 2000 National Health and Nutrition Examination Surveys, the percentages for BMI cut points were less than 23 (22.3%), 23 to less than 25 (15.09%), 25 to less than 26 (7.49%), 26 to less than 27 (7.36%), 27 to less than 28 (6.23%), 28 to less than 29 (6.30%), 29 to less than 30 (5.94%), 30 to 35 (16.95%), and more than 35 (12.62%).

We used hazard ratios reported previously¹³ to recompute annual deaths for 6 major population-based studies. The mean estimate of the total number of overweight-attributable deaths in 2000 was 494 921. For the Alameda County Health Study, the estimated number of overweight-attributable deaths in 2000 was 567 683; Framingham Heart Study, 543 981; Tecumseh Community Health Study, 462,005; American Cancer Society Cancer Pre-

Table 2. Actual Causes of Death in the United States in 1990 and 2000

Actual Cause	No. (%) in 1990*	No. (%) in 2000
Tobacco	400 000 (19)	435 000 (18.1)
Poor diet and physical inactivity	300 000 (14)	400 000 (16.6)
Alcohol consumption	100 000 (5)	85 000 (3.5)
Microbial agents	90 000 (4)	75 000 (3.1)
Toxic agents	60 000 (3)	55 000 (2.3)
Motor vehicle	25 000 (1)	43 000 (1.8)
Firearms	35 000 (2)	29 000 (1.2)
Sexual behavior	30 000 (1)	20 000 (0.8)
Illicit drug use	20 000 (<1)	17 000 (0.7)
Total	1 060 000 (50)	1 159 000 (48.2)

*Data are from McGinnis and Foege.¹ The percentages are for all deaths.

vention Study I, 451 708; Nurses Health Study, 504 602; and the National Health and Nutrition Examination Survey I Epidemiologic Follow-up Study, 439 548.¹⁴⁻¹⁹

As in the study by Allison et al, the estimate for the attributable number of deaths for nonsmokers or never-smokers was higher than the estimate for the total because smoking is associated with both lower body weight and higher mortality. Also in 2000, the mean estimate of the total number of overweight-attributable deaths among nonsmokers or never-smokers was 543 797. For the Alameda County Health Study, the estimate of overweight-attributable deaths among non-smokers or never-smokers was 639 026; Framingham Heart Study, 583 913; Tecumseh Community Health Study, 457 460; American Cancer Society Cancer Prevention Study I, 466 729; Nurses Health Study, 570 855; and the National Health and Nutrition Examination Survey I Epidemiologic Follow-up Study, 544 798. Our estimates indicate an increase of 76.6% over the 1991 estimate of overweight-attributable deaths, with more than 80% of excess deaths occurring among individuals with class 2 and 3 obesity.

The prevalence of overweight used in this study is based on data from 1999-2000. Because the effects of overweight on mortality may not appear until some years after a person becomes overweight, it is likely that the increase in prevalence of overweight in the 1990s overestimates the current actual number of deaths. However, the total num-

ber of deaths from the 1999-2000 data may well be the expected number of deaths in the next few years. Thus, we believe a more accurate and conservative estimate for overweight mortality in 2000 such as 385 000, which is the rounded average of 2000 and 1991 estimates (494 921 and 280 184).

Overweight would account for the major impact of poor diet and physical inactivity on mortality.²⁰ Diet may have a minor additional effect on mortality mainly from lack of certain essential nutrients.^{21,22} Consumption of fruits and vegetables increased in the 1990s,²³ and fat intake as a percentage of calories declined.²⁴ Physical activity has increased slightly.²⁵ We estimate that poor diet and physical inactivity will cause an additional 15 000 deaths a year, although this too may be conservative. Nutritional deficiencies alone (*International Classification of Diseases, 10th Revision [ICD-10]* codes E40-E64) were reported as the causes of 4242 deaths in 2000.

We estimate that 400 000 deaths were attributable to poor diet and physical inactivity, an increase of one third from 300 000 deaths estimated by McGinnis and Foege,¹ and the largest increase among all actual causes of death. However, poor diet and physical inactivity could account for even more deaths (>500 000) when the 1999-2000 prevalence estimates of overweight have their full effect.

Alcohol Consumption

We used 2 large nationally representative surveys to determine US alcohol

consumption. The National Health Interview Survey, a household survey that measured alcohol intake in 1999 and 2000, and the BRFSS, a telephone survey that measured alcohol intake in 1999.^{8,26}

We used RRs from the Australian National Drug and Safety Report that were based on mortality rates derived from pooled data of several studies.^{27,28} The RR values were 1.33 for hazardous drinking (4.01-6.00 drinks/d for males and 2.01-4.00 for females) and 1.47 for harmful drinking (≥ 6.01 drinks/d for males and ≥ 4.01 for females) in contrast to low levels of drinking (0.26-4.00 drinks/d for males and 0.26-2.00 for females) and abstinence (0-0.25 drinks/d for both males and females).

We used BRFSS data to compute the number of alcohol-attributable deaths for the US population aged 18 years or older. The BRFSS also asked questions about binge drinking (ie, ≥ 5 drinks per occasion). To account for the effect that respondents appeared not to include binge drinking in their reported regular drinking, we reran our analyses, adding 5 drinks per binge occasion to average drinks per day. The total number of deaths attributable to alcohol was 103 350.

We also used 3 other recent studies to estimate alcohol-attributable mortality. Two studies were based on the National Health Interview Survey^{29,30} and the National Alcohol Survey.³¹ Using all-cause mortality and RRs from these studies, we estimated approximately 60 000 deaths per year. This difference in number of deaths is mainly due to the fact that BRFSS respondents report a higher percentage of heavy drinking than do respondents in a household survey such as the National Health Interview Survey.

In another approach, we aggregated alcohol-related deaths from specified ICD codes that were summed to provide an overall estimate of deaths. In 2000, 18 539 deaths were reported as alcohol-induced (*ICD-10* codes F10, G31.2, G62.1, I42.6, K29.2, K70, R78.0, X45, X65). In addition, 16 653 persons were killed in alcohol-related crashes.³²

We estimate another 34 797 deaths in 2000 using BRFSS alcohol consumption data and disease-specific RRs from the Australian study for oropharyngeal, esophageal, liver, laryngeal, and female breast cancers; stroke; hypertensive heart disease; and other chronic liver disease and cirrhosis (ICD-10 code K73-74). This totals to 69 989 deaths in 2000 from these factors alone. In the Australian study, all-cause mortality was also higher than the summation of cause-specific mortality.

Total alcohol-attributable deaths would reach about 140 000 if mortality among previous alcohol drinkers were included. It is unclear whether excess mortality among former alcohol drinkers is due to damage or illness from past alcohol consumption.

Taking these various numbers into account, our best estimate for total alcohol-attributable deaths in 2000 is approximately 85 000, based on the conservative estimate from cause-specific deaths and the high estimate using all-cause mortality. This is a reduction of 15 000 deaths from the 1990 estimates.

Microbial Agents

We excluded human immunodeficiency virus (HIV) from this category and included it with sexual behaviors to be consistent with the analysis by McGinnis and Foege.¹ In the past, infectious agents were the leading cause of mortality.³³ These agents still present a major threat to the nation's health and are associated with high morbidity.³⁴ Several improvements in the health system have led to a decline in mortality from infectious diseases. The increase in US immunization rates led to a decline in mortality from many vaccine-preventable diseases.³⁵⁻³⁷ Several laws ensure this high immunization rate for children by requiring vaccination for school and day-care enrollment.³⁸ There also have been substantial improvements in sanitation and hygiene, antibiotics and other antimicrobial medicines, and hospital-infection control.³⁵

In 2000, influenza and pneumonia accounted for 65 313 deaths, septicemia for 31 224, and tuberculosis for

776.⁴ In general, mortality from infectious and parasitic diseases has declined since 1990.³³ Because pneumonia and septicemia occur at higher rates among patients with cancer, heart disease, lung disease, or liver disease, some of these deaths really are attributable to smoking, poor diet, and alcohol consumption.³⁹⁻⁴³ We estimate that approximately 75 000 deaths were attributable to microbial agents in 2000 from all ICD-10 codes for infectious and parasitic mortality. The major cause of the decline was a decrease in deaths from influenza and pneumonia probably reflecting at least in part an increase in immunization in older adults against vaccine-preventable diseases. This contrasts with 90 000 deaths attributed to microbial agents in 1990 estimates.

Toxic Agents

Estimating the number of deaths due to toxic agents is more challenging than any of the other risk factors due to limited published research and the challenges of measuring exposure and outcome. In the 1990s, many improvements were made in controlling and monitoring pollutants.⁴⁴ There is more systematic monitoring of pollutants at state and county levels, and exposure to asbestos, benzene, and lead have declined.⁴⁴ In fact, the US Environmental Protection Agency reported a decline of 25% from 1970 to 2001 in 6 principal air pollutants: carbon monoxide, lead, ozone, nitrogen dioxide, sulfur dioxide, and particulate matter.⁴⁵

Toxic agents are associated with increased mortality from cancer, respiratory, and cardiovascular diseases.⁴⁶⁻⁴⁹ We used the National Morbidity, Mortality, and Air Pollution Study to estimate mortality due to air pollution.⁵⁰ The study assessed the association between air pollution and mortality and morbidity in 90 cities in the United States. Only particulate matter (PM) was associated with a significant increase in mortality—an approximate 0.5% increase in total mortality for each 10- μm^3 increase in PM₁₀. Previous studies reported a range of 0.4% to 1% for that association.^{51,52} We used 23.8 μm^3 as the

daily average of PM₁₀ concentration in 2000,⁴⁵ which results in an estimate of 24 000 deaths per year (range, 22 000-52 000 deaths) from air pollution alone.

The National Institute for Occupational Safety and Health (NIOSH) estimates that about 113 000 deaths are due to occupational exposure from 1968 to 1996.⁵³ The number of deaths caused by occupational exposure has declined during that period. In 1996, NIOSH estimated 3119 deaths from pneumoconiosis and 1176 from asbestosis. Although, particulate air pollution accounts for the majority (about 60%) of mortality related to toxic agents,⁵⁴ indoor air pollution, environmental tobacco smoke, radon, lead in drinking water, and food contamination are associated with increased mortality.^{55,56} We estimate that toxic agents (excluding environmental tobacco exposure) were associated with 2% to 3.5% of total mortality in 2000. We estimate approximately 55 000 deaths attributable to toxic agents in 2000. This estimate is our least certain of the various causes.

Motor Vehicles

Motor-vehicle crashes involving passengers and pedestrians resulted in 43 354 deaths in 2000.⁴ This decline from 47 000 deaths in 1990 represents successful public health efforts in motor-vehicle safety.^{57,58} Deaths from alcohol-related crashes declined from 22 084 in 1990 to 16 653 in 2000.³² Major contributing factors include the use of child safety seats and safety belts,^{59,60} decreases in alcohol-impaired driving,⁶¹ changes in vehicle and highway design,^{62,63} and national goals to reduce motor-vehicle-related mortality and injury.⁶⁴ We estimate that approximately 26 500 deaths in 2000 were attributable to motor-vehicle crashes in which alcohol was not a factor. This is an increase of 1500 from the 1990 report because both estimates were not adjusted for the number of registered vehicles, number of crashes, nor miles of travel. We included alcohol-related deaths to stress that efforts to educate the public and enforce laws against driving while intoxicated have accounted for most of

the decline in deaths related to motor-vehicle crashes.

Firearms

Firearm-related incidents resulted in 28 663 deaths among individuals in the United States in 2000.⁴ This is a decline from approximately 36 000 deaths in 1990. The largest declines were in deaths from homicides and unintentional discharge of firearms. In 2000, 16 586 deaths were due to intentional self-harm (suicide) by discharge of firearms (*ICD-10* codes X72-X74). Assault (homicide) by discharge of firearms (*ICD-10* codes X93-X95) resulted in 10 801 deaths. Unintentional discharge of firearms (*ICD-10* codes W32-W34) resulted in 776 deaths, while discharge of firearms, undetermined intent (*ICD-10* codes Y22-Y24), resulted in 230 deaths. The remaining 270 deaths were due to legal intervention (*ICD-10* code Y35). These numbers were ascertained from death certificate reports.

Sexual Behavior

Sexual behavior is associated with an increased risk of preventable disease and disability.⁶⁵ An estimated 20 million persons are newly infected with sexually transmitted diseases each year in the United States.^{66,67} Mortality from sexually transmitted diseases is declining due to the availability of earlier and better treatment, especially for HIV.^{67,68} In 2000, HIV disease (*ICD-10* codes B20-B24) resulted in 14 578 deaths. In 1990, HIV was the cause of 27 695 deaths for persons older than 13 years, indicating about a 48% decline in HIV mortality during the decade. Based on the sexual behavior-attributable fraction from the literature,⁶⁹⁻⁷¹ we estimate that 20 000 deaths (range, 18 000-25 000 deaths) in 2000 were due to sexual behavior—mainly HIV; other contributors were hepatitis B and C viruses and cervical cancer. The decline of 10 000 deaths from the 1990 estimates¹ was due to the decline in HIV mortality.

Illicit Use of Drugs

Illicit drug use is associated with suicide, homicide, motor-vehicle injury,

HIV infection, pneumonia, violence, mental illness, and hepatitis.^{27,28,72-77} An estimated 3 million individuals in the United States have serious drug problems.^{78,79} Several studies have reported an undercount of the number of deaths attributed to drugs by vital statistics⁸⁰; however, improved medical treatments have reduced mortality from many diseases associated with illicit drug use. In keeping with the report by McGinnis and Foege,¹ we included deaths caused indirectly by illicit drug use in this category. We used attributable fractions to compute the number of deaths due to illicit drug use.^{27,28,81} Overall, we estimate that illicit drug use resulted in approximately 17 000 deaths in 2000, a reduction of 3000 deaths from the 1990 report.

Other Factors

Several other factors contribute to an increased rate of death. There are factors that we do not know of such as unknown pollutants or perhaps exposures that may cause a considerable number of deaths. Poverty and low education levels are associated with increased mortality from many causes,^{82,83} partly due to differential exposure to the risks described above. However, controlling for differential exposure to risk factors is unlikely to explain the entire impact on mortality. Lack of access to proper medical care or preventive services is associated with increased mortality.⁸⁴ Biological characteristics and genetic factors also greatly affect risk of death.⁸⁵ In most studies we reviewed, low education levels and income were associated with increased risk of cardiovascular disease, cancer, diabetes, and injury. The Healthy People 2010 initiative has made the elimination of health disparities, especially racial and ethnic disparities, a primary goal.⁸⁶

COMMENT

We found that about half of all deaths that occurred in the United States in 2000 could be attributed to a limited number of largely preventable behaviors and exposures. Overall, we found

relatively minor changes from 1990 to 2000 in the estimated number of deaths due to actual causes. Our findings indicate that interventions to prevent and increase cessation of smoking, improve diet, and increase physical activity must become much higher priorities in the public health and health care systems.

The most striking finding was the substantial increase in the number of estimated deaths attributable to poor diet and physical inactivity. We estimate that roughly 400 000 deaths now occur annually due to poor diet and physical inactivity. The gap between deaths due to poor diet and physical inactivity and those due to smoking has narrowed substantially. Because rates of overweight increased rapidly during the 1990s, we used a conservative approach to make our estimates, accounting for the delayed effects of overweight on mortality. In addition, overweight lessens life expectancy.^{87,88} However, it is clear that if the increasing trend of overweight is not reversed over the next few years, poor diet and physical inactivity will likely overtake tobacco as the leading preventable cause of mortality.

The most disappointing finding may be the slow progress in reducing tobacco-related mortality. A few states, notably California, have had major success in programs that led to reducing deaths from heart disease and cancer.⁸⁹ However, efforts in most other states are too recent or short-term to have a similar effect. In response to the increase in tobacco use among youth in the early 1990s, state and national tobacco-control efforts increased their focus on prevention of initiation and recognized the importance of cessation on reducing smoking-related deaths. Thus, most national and state efforts now address comprehensive program strategies.⁹⁰ Current tobacco-control efforts will also need strong cessation components to show a decline in tobacco deaths in a future assessment. Recent reports on the effects of telephone quit lines for smokers are encouraging.⁹¹ On the other hand, large

state budget shortfalls are leading to large cuts in public health, with a corresponding diversion of resources from tobacco taxes and settlement dollars to cover deficits instead of tobacco-control programs.

Despite the call to action on these risk factors a decade ago, there has been little progress in reducing the total number of deaths from these causes. The progress that has occurred primarily involves actual causes of death that are less prominent. With the shift in the age distribution of the population, more adults now are in the age group at highest risk because of the cumulative effects of their behavior. The net effect is that both total deaths and total burden due to the actual causes have increased.

Our analyses have several limitations. Our study reported actual causes of mortality in the United States. However, these causes are also associated with a large morbidity burden. In addition to premature death, years of lost life, diminished productivity, and high rates of disability, decreased quality of life is also strongly associated with these actual causes. A recent World Health Organization report finds these actual causes of death to be the leading causes of total disease burden, not just mortality, in the developed world.⁹² Because we used self-reported estimates for some risk behaviors, (ie, prevalence of alcohol intake) they may have been underestimated. Finally, using all-cause mortality may result in overestimates of the number of deaths from specific causes. In addition, if the effect of the risk factor is age-dependent, then age- and sex-specific estimates are preferable.

Our analyses did not assess the effect of genetics. Genetic factors have been associated with several diseases discussed herein.⁸⁵ Much of the impact of genetics is likely mediated through increased physical susceptibility to these behavioral and other modifiable risks. However, increases in obesity and diabetes cannot be due to widespread changes in the human genome over the last 10 years. Nevertheless, genetics of-

fers great potential for treating and ameliorating risk. Identifying individuals at higher risk for a disease through genetic testing may promote lifestyle changes that can help prevent the onset of that disease.⁹³

In this study we also did not examine the effects of high blood pressure and cholesterol or lipid profile on mortality, although some of the effects of these factors are mediated through poor diet and physical inactivity. These risk factors are common among adults in the United States. More than 30% of US adults have high blood pressure or high cholesterol.^{94,95} Monitoring and controlling blood pressure and cholesterol is crucial to preventing premature mortality and morbidity.

One of the most difficult aspects of this analysis is that the attribution of the actual cause that led to death varies depending on perspective. We used similar methods to those used by McGinnis and Foege¹ to allow comparisons. We tried when possible to use RRs that are fully adjusted for other risk factors in our analyses, but possibly not eliminating duplicate attribution of causes. We also explicitly included some deaths in more than 1 category (eg, alcohol and motor vehicle crashes) when choosing another category seemed as though it might artificially constrain interpretation for future prevention programs.

In summary, smoking and the deaths attributed to the constellation of poor diet and physical inactivity currently account for about one third of all deaths in the United States. The rapid increase in the prevalence of overweight means that this proportion is likely to increase substantially in the next few years. The burden of chronic diseases is compounded by the aging effects of the baby boomer generation and the concomitant increased cost of illness at a time when health care spending continues to outstrip growth in the gross domestic product of the United States. In ancient times, Hippocrates stated that "the function of protecting and developing health must rank even above that of restoring it when it is impaired." The

findings in this study argue persuasively for the need to establish a more preventive orientation in health care and public health systems in the United States.

Author Contributions: Dr Mokdad had full access to the data in this study and takes full responsibility for the scientific integrity of the data and the accuracy of the analysis and content of the manuscript.

Study concept and design: Mokdad, Marks, Stroup, Gerberding.

Acquisition of data: Mokdad, Stroup, Gerberding.

Analysis and interpretation of data: Mokdad, Marks, Stroup, Gerberding.

Drafting of the manuscript: Mokdad, Marks, Stroup, Gerberding.

Critical revision of the manuscript for important intellectual content: Mokdad, Marks, Stroup, Gerberding.

Statistical expertise: Mokdad, Stroup.

Obtained funding: Marks, Gerberding.

Administrative, technical, or material support: Mokdad, Marks, Stroup, Gerberding.

Study supervision: Marks, Gerberding.

Funding/Support: There was no external funding for this work.

Acknowledgment: We acknowledge the valuable contributions of Barbara A. Bowman, PhD, Robert D. Brewer, MD, MSPH, Earl S. Ford, MD, MPH, Wayne H. Giles, MD, James M. Mendlein, PhD, Cheryl Pellerin, Susan Y. Chu, PhD, and Eduardo J. Simoes, MD, MPH.

Role of the Sponsor: The Centers for Disease Control and Prevention reviewed and approved this report before submission.

REFERENCES

- McGinnis JM, Foege WH. Actual causes of death in the United States. *JAMA*. 1993;270:2207-2212.
- Health, United States 2002. Rockville, Md: Dept of Health and Human Services, Centers for Disease Control and Prevention; 2002. DHHS Publication No. 1232.
- Koplan JP, Dietz WH. Caloric imbalance and public health policy. *JAMA*. 1999;282:1579-1581.
- Minino AM, Arias E, Kochanek KD, Murphy SL, Smith BL. Deaths: final data for 2000. *Natl Vital Stat Rep*. 2002;50:1-120.
- Annual smoking attributable mortality, years of potential life lost and economic costs: United States 1995-1999. *MMWR Morb Mortal Wkly Rep*. 2002;51:300-303.
- Centers for Disease Control and Prevention. Smoking-attributable mortality, morbidity, and economic costs. Available at: <http://www.cdc.gov/tobacco/sammec>. Accessibility verified February 10, 2004.
- Thun MJ, Day-Lally C, Myers DG, et al. Trends in tobacco smoking and mortality from cigarette use in Cancer Prevention Studies I (1959 through 1965) and II (1982 through 1988). In: *Changes in Cigarette-Related Disease Risks and Their Implication for Prevention and Control: Smoking and Tobacco Control Monograph 8*. Bethesda, Md: US Dept of Health and Human Services, Public Health Service, National Institutes of Health, National Cancer Institute; 1997: 305-382.
- Mokdad AH, Stroup DF, Giles WH. Public health surveillance for behavioral risk factors in a changing environment: recommendations from the Behavioral Risk Factor Surveillance Team. *MMWR Recomm Rep*. 2003;52:(RR-9)1-12.
- Clinical Guidelines on the Identification, Evaluation, and Treatment of Overweight and Obesity in Adults: the Evidence Report. Rockville, Md: National

Institutes of Health, National Heart, Lung, and Blood Institute; 1998.

10. Mokdad AH, Bowman BA, Ford ES, Vinicor F, Marks JS, Koplan JP. The continuing epidemics of obesity and diabetes in the US. *JAMA*. 2001;286:1195-1200.
11. Flegal KM, Carroll MD, Ogden CL, Johnson CL. Prevalence and trends in obesity among US adults, 1999-2000. *JAMA*. 2002;288:1723-1727.
12. Centers for Disease Control and Prevention. Plan and operation of the Third National Health and Nutrition Examination Survey, 1988-94. *Vital Health Stat*. 1994;1:1-307.
13. Allison DB, Fontaine KR, Manson JE, Stevens J, Van Itallie TB. Annual deaths attributable to obesity in the United States. *JAMA*. 1999;282:1530-1538.
14. Berkham LF, Breslow L. *Health and Ways of Living: the Alameda County Studies*. New York, NY: Oxford University Press; 1983.
15. Dawber TR, Meadors GF, Moore FE. Epidemiological approaches to heart disease: the Framingham Study. *Am J Public Health*. 1951;41:279-286.
16. Epstein FH, Napier JA, Block WD, et al. The Tecumseh Study: design, progress, and perspectives. *Arch Environ Health*. 1970;21:402-407.
17. Lew EA, Garfinkel L. Variations in mortality by weight among 750,000 men and women. *J Chronic Dis*. 1979;32:563-576.
18. Stampfer MJ, Willett WC, Colditz GA, Rosner B, Speizer FE, Hennekens CH. A prospective study of postmenopausal estrogen therapy and coronary heart disease. *N Engl J Med*. 1985;313:1044-1049.
19. Cox CS, Mussolino M, Rothwell ST, et al. Plan and operation of the NHANES I Epidemiologic Follow-up Study, 1992. *Vital Health Stat*. 1997;35:1-231.
20. Blair SN, Nichaman MZ. The public health problem of increasing prevalence rates of obesity and what should be done about it. *Mayo Clin Proc*. 2002;77:109-113.
21. Hu FB, Willett WC. Optimal diets for prevention of coronary heart disease. *JAMA*. 2002;288:2569-2578.
22. Peto J. Cancer epidemiology in the last century and the next decade. *Nature*. 2001;411:390-395.
23. Ruowei L, Serdula M, Bland S, Mokdad A, Bowman B, Nelson D. Trends in fruit and vegetable consumption among adults in 16 US states: Behavioral Risk Factor Surveillance System 1990-1996. *Am J Public Health*. 2000;90:777-781.
24. Ernst ND, Sempos CT, Briefel RR, Clark MB. Consistency between US dietary fat intake and serum cholesterol concentrations: the National Health and Nutrition Examination Surveys. *Am J Clin Nutr*. 1997;66:965S-972S.
25. Physical activity trends—United States, 1990-1998. *MMWR Morb Mortal Wkly Rep*. 2001;50:166-169.
26. Centers for Disease Control and Prevention, National Center for Health Statistics. National Health Interview Survey. Available at: <http://www.cdc.gov/nchs>. Accessibility verified February 10, 2004.
27. Stevenson RB. *The Quantification of Drug-Caused Mortality and Morbidity in Australia, 1998*. Canberra: Australian Institute of Health and Welfare, Commonwealth Department of Human Services and Health; 2001. Category No. PHE 29.
28. English DR, Holman CDJ, Milne E, et al. *The Quantification of Drug-Caused Morbidity and Mortality in Australia, 1995*. Canberra: Commonwealth Department of Human Services and Health; 1995.
29. Liao Y, McGee DL, Cao G, Cooper RS. Alcohol mortality: findings from the National Health Interview Survey (1988 and 1990). *Am J Epidemiol*. 2000;151:651-659.
30. Dawson DA. Alcohol consumption, alcohol dependence, and all-cause mortality. *Alcohol Clin Exp Res*. 2000;24:72-81.
31. Rehm J, Greenfield TK, Rogers JD. Average volume of alcohol consumption, patterns of drinking, and all-cause mortality: results from the US national alcohol survey. *Am J Epidemiol*. 2001;153:64-71.
32. National Highway Traffic Safety Administration. *Traffic Safety Fact, 2000*. Washington, DC: US Dept of Transportation; 2001.
33. Armstrong GL, Conn LA, Pinner RW. Trends in infectious disease mortality in the United States during the 20th century. *JAMA*. 1999;281:61-66.
34. Simonsen L, Conn LA, Winner RW, Teutsch S. Trends in infectious disease hospitalizations in the United States, 1980-1994. *Arch Intern Med*. 1998;158:1923-1928.
35. Control of infectious disease. *MMWR Morb Mortal Wkly Rep*. 1999;48:621-629.
36. Status report on Childhood Immunization Initiative: reported cases of selected vaccine-preventable diseases—United States, 1996. *MMWR Morb Mortal Wkly Rep*. 1997;46:665-671.
37. Influenza and pneumococcal vaccination levels among persons aged ≥ 65 years—United States, 2001. *MMWR Morb Mortal Wkly Rep*. 2002;51:1019-1024.
38. Vaccination coverage among children enrolled in Head Start programs and licensed child care centers and entering school—United States and selected reporting areas, 1999-2000 school year. *MMWR Morb Mortal Wkly Rep*. 2001;50:847-855.
39. Ortvist A, Kalin M, Julander I, Mufson MA. Deaths in bacteremic pneumococcal pneumonia: a comparison of two populations—Huntington, W VA, and Stockholm, Sweden. *Chest*. 1993;103:710-716.
40. Valdez R, Narayam KM, Geiss L, Engelgau MM. Impact of diabetes mellitus on mortality associated with pneumonia and influenza among non-Hispanic black and white US adults. *Am J Public Health*. 1999;89:1715-1721.
41. Koziel H, Koziel MJ. Pulmonary complications of diabetes mellitus: pneumonia. *Infect Dis Clin North Am*. 1995;9:65-96.
42. Simonsen L, Clarke MJ, Williamson GD, Stroup DF, Arden NH, Schonberger LB. The impact of influenza epidemics on mortality: introducing a severity index. *Am J Public Health*. 1997;87:1944-1950.
43. Simonsen L, Clarke MJ, Stroup DF, Williamson GD, Arden NH, Cox NJ. A method for timely assessment of influenza-associated mortality in the United States. *Epidemiology*. 1997;8:390-395.
44. *National Air Quality and Emissions Trends Report, 1999*. Research Triangle Park, NC: US Environmental Protection Agency, Office of Air Quality Planning and Standards; 2001.
45. US Environmental Protection Agency. Air trends—six principal pollutants. Available at: <http://www.epa.gov/airtrends/sixpoll.html>. Accessibility verified February 10, 2004.
46. Styper P, McMillan N, Gao F, Davis J, Sacks J. Effect of outdoor airborne particulate matter on daily death counts. *Environ Health Perspect*. 1995;103:490-497.
47. Burnett RT, Cakmak S, Brook JR. The effect of the urban air pollution mix on daily mortality rates in 11 Canadian cities. *Can J Public Health*. 1998;89:152-156.
48. Beckett WS. Current concepts: occupational respiratory diseases. *N Engl J Med*. 2000;342:406-413.
49. Peto J. Cancer epidemiology in the last century and the next decade. *Nature*. 2001;411:390-395.
50. Samet JM, Zeger SL, Dominici F, et al. *The National Morbidity, Mortality and Air Pollution Study, Part II: Morbidity and Mortality From Air Pollution in the United States*. Cambridge, Mass: Health Effects Institute; 2000. Research report 94.
51. Katsouyanni K, Touloumi G, Spix C, et al. Short-term effects of ambient sulphur dioxide and particulate matter on mortality in 12 European cities: results from time series data from the Air Pollution and Health: A European Approach project. *BMJ*. 1997;314:1658-1663.
52. Pope CA III, Dockery DW, Schwartz J. Review of epidemiological evidence of health and effects of particulate air pollution. *Inhal Toxicol*. 1995;7:1-18.
53. National Institute for Occupational Safety and Health. Worker health chartbook, 2000. Available at: <http://www.cdc.gov/niosh/00-127pd.html>. Accessibility verified February 10, 2004.
54. de Hollander AE, Melse JM, Lebre E, Kramers PG. An aggregate public health indicator to represent the impact of multiple environmental exposures. *Epidemiology*. 1999;10:606-617.
55. Khan AS, Swerdlow DL, Juranek DD. Precautions against biological and chemical terrorism directed at food and water supplies. *Public Health Rep*. 2001;116:3-14.
56. Smith KR, Corvalan CF, Kjellstrom T. How much global ill health is attributable to environmental factors? *Epidemiology*. 1999;10:573-584.
57. Motor vehicle safety: a 20th century public health achievement [published correction appears in *MMWR Morb Mortal Wkly Rep*. 1999;48:473]. *MMWR Morb Mortal Wkly Rep*. 1999;48:369-374.
58. Task Force on Community Prevention Services. Motor-vehicle occupant injury: strategies for increasing use of child safety seats, increasing use of safety belts, and reducing alcohol-impaired driving. *MMWR Recomm Rep*. 2001;50(RR-7):1-14.
59. *Traffic Safety Facts 1999: Occupant Protection*. Washington, DC: US Dept of Transportation, National Highway Traffic Safety Administration; 2000. Publication No. DOT HS 809090.
60. Zador PL, Krawchuk SA, Voas RB. Alcohol-related relative risk of driver fatalities and driver involvement in fatal crashes in relation to driver age and gender: an update using 1996 data. *J Stud Alcohol*. 2000;61:387-395.
61. Zaza S, Wright-De Agüero LK, Briss PA, et al. Data collection instrument and procedure for systematic reviews in the guide to community preventive services. *Am J Prev Med*. 2000;18:44-74.
62. Transportation Research Board. *Safety Research for a Changing Highway Environment*. Washington, DC: National Research Council, Transportation Research Board; 1990. Special report No. 229.
63. Centers for Disease Control and Prevention and National Highway Traffic Safety Administration. *Position Papers From the Third National Injury Control Conference: Setting the National Agenda for Injury Control in the 1990s*. Washington, DC: US Dept of Health and Human Services, Public Health Service, Centers for Disease Control and Prevention; 1992.
64. US Department of Health and Human Services. *Healthy People 2010: Understanding and Improving Health and Objectives for Improving Health*. Washington, DC: US Government Printing Office; 2000.
65. Murry CJL, Lopez AD, eds. *Health Dimensions of Sex and Reproduction: the Global Burden of Sexually Transmitted Diseases, HIV, Maternal Conditions, Perinatal Disorders, and Congenital Anomalies*. Geneva, Switzerland: World Health Organization; 1998.
66. Michaud CM, Murray CJL, Bloom BR. Burden of diseases: implications for future research. *JAMA*. 2001;285:535-539.
67. Centers for Disease Control and Prevention. *Sexually Transmitted Disease Surveillance 2000*. Atlanta, Ga: US Dept of Health and Human Services, Centers for Disease Control and Prevention; 2001.
68. Patella FJ, Delaney KM, Moorman AC, et al. Declining morbidity and mortality among patients with advanced human immunodeficiency virus infection. *N Engl J Med*. 1998;338:853-860.
69. Ebrahim SH, Peterman TA, Zaidi AA, Kamb ML. Mortality related to sexually transmitted diseases in US women, 1973 through 1992. *Am J Public Health*. 1997;87:938-944.
70. Cates W. Estimates of the incidence and prevalence of sexually transmitted diseases in the United States. *Sex Transm Dis*. 1999;26:S2-S7.

71. Recommendations for prevention and control of hepatitis C virus (HCV) infection and HCV-related chronic disease. *MMWR Morb Mortal Wkly Rep.* 1998;47:1-39.
72. Brook DW, Brook JS, Zhang C, Cohen P, White-man M. Drug use and the risk of major depressive disorder, alcohol dependence, and substance use disorders. *Arch Gen Psychiatry.* 2002;59:1039-1044.
73. Volkow ND. Drug abuse and mental illness: progress in understanding comorbidity. *Am J Psychiatry.* 2001;158:1181-1183.
74. Oyefeso A, Ghodse H, Clancy C, Corkery J, Gold-finch R. Drug abuse-related mortality: a study of teen-age addicts over a 20-year period. *Soc Psychiatry Psychi- atr Epidemiol.* 1999;34:437-441.
75. Phillips DP, Christenfeld N, Ryan NM. An in- crease in the number of deaths in the United States in the first week of the month—an association with substance abuse and other causes of death. *N Engl J Med.* 1999;341:93-98.
76. Rivara FP, Mueller BA, Somes G, Mendoza CT, Rushforth NB, Kellermann AL. Alcohol and illicit drug abuse and the risk of violent death in the home. *JAMA.* 1997;278:569-575.
77. Bruner AB, Fishman M. Adolescents and illicit drug use. *JAMA.* 1998;280:597-598.
78. Pope HG Jr, Ionescu-Pioggia M, Pope KW. Drug use and life style among college undergraduates: a 30-year longitudinal study. *Am J Psychiatry.* 2001;158: 1519-1521.
79. Johnston LD, O'Malley PM, Bachman JG. *National Survey Results on Drug Use From the Moni- toring the Future study, 1975-1999. Volume I: Sec- ondary School Students.* Ann Arbor: University of Michigan; 2000.
80. Pollack DA, Holmgren P, Lui K, Kirk ML. Dis- crepancies in the reported frequency of cocaine deaths, United States, 1983 through 1988. *JAMA.* 1991;266: 2233-2237.
81. Single E, Rehm J, Robson L, Truong MV. The rela- tive risks and etiologic fractions of different causes of death and disease attributable to alcohol, tobacco and illicit drug use in Canada. *CMAJ.* 2000;162:1669- 1675.
82. Rogot E, Sorlie PD, Johnson NJ. Life expectancy by employment status, income, and education in the National Longitudinal Mortality Study. *Public Health Rep.* 1992;107:457-461.
83. Wong MD, Shapiro MF, Boscardin WJ, Ettner SL. Contribution of major diseases to disparities in mor- tality. *N Engl J Med.* 2002;347:1585-1592.
84. Franks P, Clancy CM, Gold MR. Health insur- ance and mortality: evidence from a national cohort. *JAMA.* 1993;270:737-741.
85. Khoury MJ, Beaty TH, Cohen BH. *Fundamentals of Genetic Epidemiology.* New York, NY: Oxford Uni- versity Press; 1993.
86. Department of Health and Human Services. *Healthy People 2010: Understanding and Improving Health.* Washington, DC: Government Printing Of- fice; 2000.
87. Fontaine KR, Redden DT, Wang C, Westfall AO, Allison DB. Years of life lost due to obesity. *JAMA.* 2003;289:187-193.
88. Peeters A, Barendregt JJ, Willekens F, et al, for the Netherlands Epidemiology and Demography Compression of Morbidity Research Group. Obesity in adulthood and its consequences for life expec- tancy: a life-table analysis. *Ann Intern Med.* 2003; 138:24-32.
89. Centers for Disease Control and Prevention. De- clines in lung cancer rates—California, 1988-1997. *MMWR Morb Mortal Wkly Rep.* 2000;49:1066- 1069.
90. Reducing tobacco use: a report of the Surgeon General. *MMWR Recomm Rep.* 2000;49(RR-16): 1-27.
91. Zhu SH, Anderson CM, Tedeschi GJ, et al. Evi- dence of real-world effectiveness of a telephone quit- line for smokers. *N Engl J Med.* 2002;347:1087- 1093.
92. Ezzati M, Lopez AD, Rodgers A, Vander Hoorn S, Murray CJ, and the Comparative Risk Assessment Collaborating Group. Selected major risk factors and global and regional burden of disease. *Lancet.* 2002; 360:1347-1360.
93. Bruke W. Genetic testing. *N Engl J Med.* 2002; 347:1867-1875.
94. Ford ES, Mokdad AH, Giles WH, Mensah GA. Se- rum total cholesterol concentrations and awareness, treatment, and control of hypercholesterolemia among US adults: findings from the National Health and Nu- trition Examination Survey 1999-2000. *Circulation.* 2003;107:2185-2189.
95. Hajjar I, Kotchen TA. Trends in prevalence, aware- ness, treatment, and control of hypertension in the United States, 1988-2000. *JAMA.* 2003;290:199- 206.
96. Strum R. The effects of obesity, smoking, and prob- lem drinking on chronic medical problems and health care costs. *Health Aff (Millwood).* 2002;21:245- 253.
97. Visscher TL, Seidell JC. The public health impact of obesity. *Annu Rev Public Health.* 2001;22: 355-375.

that it may not be correct to assume that methadone is a much safer and less abusable alternative to other opiate analgesics.

Theodore J. Cicero, PhD
cicerot@wustl.edu

Washington University School of Medicine
St Louis, Mo

James A. Inciardi, PhD
University of Delaware
Newark

Financial Disclosure: The authors received consultant fees from Purdue Pharma LLC during the time these studies were carried out.

Access to Data: Dr Cicero had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analyses.

Funding/Support: This research was supported in part by grants from Purdue Pharma LLC as a general study of prescription drug abuse. Purdue Pharma has extensive interests in opiate analgesics, the most important of which is oxycodone, and other long-acting analgesic preparations.

Role of Sponsors: Purdue Pharma LLC had no role in the design and conduct of the study; the collection, analysis, and interpretation of the data; or the preparation, review, or approval of the manuscript.

1. Executive Office of the President; Office of National Drug Control Policy. *Pulse Check: Trends in Drug Abuse, January 2004*. Washington, DC: Office of National Drug Control Policy; 2004.
2. National Drug Intelligence Center. Information Bulletin: Methadone Abuse Increasing. September 2003. Available at: <http://www.usdoj.gov/ndic/pubs6/6292/index.htm>. Accessibility verified November 23, 2004.
3. Substance Abuse and Mental Health Services Administration. *Methadone-Associated Mortality: Report of a National Assessment*. Rockville, Md: Center for

Substance Abuse Treatment, Substance Abuse and Mental Health Services Administration; 2004. Publication 04-3904.

4. Substance Abuse and Mental Health Services Administration. *Results From the 2002 National Survey on Drug Use and Health: National Findings*. Rockville, Md: Substance Abuse and Mental Health Services Administration; 2003. NHSDA Series H22, DHSS publication SMA 03-3836.

5. Maimon A. Methadone abuse hits state hard. *Louisville Courier Journal*. May 9, 2004.

6. Cicero TJ, Adams EH, Geller A, et al. A postmarketing surveillance program to monitor tramadol (tramadol hydrochloride) abuse in the United States. *Drug Alcohol Depend*. 1999;57:7-22.

CORRECTION

Errors in Data Reporting: In the Special Communication entitled "Actual Causes of Death in the United States, 2000" published in the March 10, 2004, issue of THE JOURNAL (2004;291:1238-1245), there were multiple errors in reported data. On page 1238 in the Results section of the Abstract, "(400 000 deaths; 16.6%)" should be "(365 000 deaths; 15.2%)." On page 1239, in the third column, in the first paragraph, "23 (22.3%)" should be "23 (22.03%)." On page 1240, in Table 2, "400 000 (16.6)" deaths for "poor diet and physical inactivity" in 2000 should be "365 000 (15.2)." A dagger symbol should be added to "alcohol consumption" in the body of the table and a dagger footnote should be added with "In 1990 data, deaths from alcohol-related crashes are included in alcohol consumption deaths, but not in motor vehicle deaths. In 2000 data, 16 653 deaths from alcohol-related crashes are included in both alcohol consumption and motor vehicle death categories." Also on page 1240, third column, fifth paragraph, "18 539 deaths" should be "19 358 deaths." On page 1241, first column, first paragraph, "69 989 deaths" should be "70 808 deaths." On page 1242, first column, first paragraph, "remaining 270 deaths" should be "remaining 359 deaths" and in the second paragraph, "resulted in 14 578 deaths" should be "resulted in 14 478 deaths." Also on page 1242, third column, second paragraph, "roughly 400 000 deaths" should be "roughly 365 000 deaths."

Call for Submissions: Archives of Internal Medicine

The editors of the *Archives of Internal Medicine* seek artistic photographs or photographs of artwork done by *Archives* readers for reproduction on the journal's cover. Submissions must be the author's own work; work that has to do with the themes of medicine is of particular interest. Sculpture, paintings, drawings, photography, fabric art, graphic art, metalwork, crafts, computer art, depictions of medical specimens—perhaps herbs or historical artifacts—and other forms of art are acceptable as long as they can be captured in a photographic submission. No recognizable persons should appear in the image. The image may be black and white or color and at least 3.5×5 in (7.6×12.7 cm) and no larger than 8×10 in (20.3×25.4 cm). See Instructions for Authors at the *Archives*' Web site (<http://www.archinternmed.com>) for submission information. Text of fewer than 250 words written by the artist about the work should accompany submissions.